
CAPSTONE TECHNICAL DELIVERY

UVA STAT 4996 CAPSTONE PROJECT

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1 Introduction

Our research aims to utilize statistical machine learning models to predict the Health Opportunity Index (HOI) in Virginia, taking into account various influencing factors including Social Determinants of Health. Then, we will evaluate and identify the most effective model for this prediction. Furthermore, we intend to expand the impact of our analysis beyond Virginia to other states in the United States, using the selected model.

Predicting the Health Opportunity Index (HOI) holds significant importance as it provides a data-driven approach to exploring the relationship between health opportunities and various socioeconomic, environmental, and healthcare factors. This predictive capability enables targeted interventions and prioritization of investments, resources, and programs where they are most needed to address health disparities.

We aim to provide a systematic approach to quantitatively predict an area's health opportunities. It enables the identification of regions with the greatest need for support and intervention and facilitates the development of holistic policy solutions by utilizing resources to address Social Determinants of Health.

By accurately predicting HOI, decision-makers, leaders, and advocates gain valuable insights into the neighborhoods most disproportionately affected by these disparities. Thus, our analysis findings empower policymakers to make informed decisions on where to direct funding, healthcare services, education programs, and community development initiatives.

2 Research Question

How does the Statistical Machine Learning Model predict the Health Opportunity Index (HOI) in Virginia and other states based on Influencing Indicators?

3 Data Collection and Preparation

Here are the datasets we will use in this study:

Dataset Name	Source of Collection	Description
VDH_VA_HOI	VDH collected	Health Opportunity Index Dataset for Virginia
ACS_VA	Team Terminators collected	2013-2017 ACS Virginia raw data used for machine learning models
ACS_CA	Team Terminators collected	2013-2017 ACS California raw data for model application

Table 1: Summary of Datasets Used in the Study

To address our research question regarding the prediction of the HOI using statistical machine learning models, we sourced and processed several datasets. The HOI is a group of indicators that provide broad insight into the overall opportunity Virginians have to live long and healthy lives based on the Social Determinants of Health (SDH). SDH are the non-medical factors that influence health outcomes. They are the conditions in which people are born, grow, work, live, and age, and the wider set of forces and systems shaping the conditions of daily life. There are variables in

different fields such as demographics (e.g., age, sex), and socioeconomic (household income, education). HOI is a hierarchical index that allows users to examine social determinants of health at multiple levels of detail in Virginia. It is made up of over 30 variables, combined into 13 indicators, grouped into four profiles, which are aggregated into a single Health Opportunity Index.

Our initial dataset, the Virginia Health Opportunity Index (VDH_VA_HOI), was obtained from the Virginia Department of Health (VDH). The VDH is dedicated to protecting and promoting the health of Virginians. This dataset is constructed from the 2013-2017 American Community Survey (ACS) 5-year estimates, reflecting a comprehensive compilation of social determinants of health across Virginia. The ACS is an ongoing survey that provides vital information on a yearly basis about our nation and its people. We define that dataset as VDH_VA_HOI dataset. To understand the construction of the VDH_VA_HOI, we engaged with the HOI data team at VDH and reviewed the relevant documentation and methodologies provided by the team. We gained insight into the structure and methodology of the Virginia Health Opportunity Index. We learned about the Index's comprehensive framework, which categorizes more than 30 variables into 13 indicators across four dimensions that reflect a wide range of social determinants of health. For example, the Community Environmental Profile includes indicators such as air quality and walkability, which measure both environmental factors and the impact of community infrastructure on health.

With insights into the VDH methodology, we replicated the data collection process using the Social Explorer website, which is a website provides easy access to demographic information about the United States including ACS data, to gather the 2013-2017 5-year estimate ACS raw data for Virginia. This self-collected dataset is referred to as the ACS_VA dataset. It includes selected features aligned with the official HOI documentation, focusing on key social determinants of health. Our goal was to emulate VDH's approach and create a HOI dataset with strong predictive power.

Here are the features being selected:

```
===== Social Explorer Tables: ACS 2017 (5-Year Estimates) =====
A00002. Population Density (Per Sq. Mile)
A00001. Total Population
A10003. Average Household Size
A10003B. Average Household Size of Renter-Occupied Housing Units
A12004. School Enrollment for the Population 3 Years and Over
A12003. School Dropout Rate for Population 16 to 19 Years
A17005. Unemployment Rate for Civilian Population in Labor Force 16 Years and Over
A14001. Household Income (In 2017 Inflation Adjusted Dollars)
A14010. Median Family Income (In 2017 Inflation Adjusted Dollars)
A14028. Gini Index of Income Inequality
A10038. Monthly Housing Cost (Renter-Occupied Housing Units)
A17002. Employment Status for Total Population 16 Years and Over
```

```
A17002. Employment Status for Total Population 16 Years and Over
A01003B. Age of Householder (Renter-Occupied Housing Units)
A10038B. Monthly Housing Costs as a Percentage of Household Income in the Past 12 Months (Renter-Occupied Housing Units)
A13002. Poverty Status in 2017 of Families by Family Type by Presence of Children Under 18 Years
A13003A. Poverty Status in 2017 for Children Under 18
A13003B. Poverty Status in 2017 for Population Age 18 to 64
A13003C. Poverty Status in 2017 for Population Age 65 and Over
A13004. Ratio of Income in 2017 to Poverty Level
A20001. Health Insurance
```

Figure 2: Feature Selection Part 2

Figure 1: Feature Selection Part 1

The ACS_VA dataset underwent rigorous data cleaning and preprocessing to ensure quality and relevance. We removed irrelevant features, such as redundant geographical identifiers and variables with no predictive power regarding health opportunities. For instance, attributes such as "Area (Land).1" were excluded promptly as their relationship to health opportunities is non-essential and potentially confounding. We then selected features based on their importance as outlined in the VDH methodology, ensuring each variable directly contributes to understanding or predicting the HOI.

We then embarked on a strategic consolidation of variables to construct more potent indicators. A prime example of this is the aggregation of income bracket variables into a singular indicator, "Percentage of Households: Below \$30,000". This indicator was crafted by summing the percentages of households within the lowest income brackets and then discarding the disaggregated components post-calculation. The rationale for this aggregation was twofold: to reduce dimensionality, simplifying our model without sacrificing detail, and to create an indicator that encapsulates a broader spectrum of economic vulnerability which is a significant determinant of health opportunity.

```
## Create a new variable for the percentage of households below $30,000
## This variable is created by summing the percentages of households in the lowest income brackets
## and then discarding the disaggregated components post-calculation.
## The rationale for this aggregation was twofold: to reduce dimensionality, simplifying our model without sacrificing detail,
## and to create an indicator that encapsulates a broader spectrum of economic vulnerability which is a significant determinant of health opportunity.

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# and then discarding the disaggregated components post-calculation.
# The rationale for this aggregation was twofold: to reduce dimensionality, simplifying our model without sacrificing detail,
# and to create an indicator that encapsulates a broader spectrum of economic vulnerability which is a significant determinant of health opportunity.
```

Figure 3: Combine Features

```
## Drop unrelated features
## These features are dropped as they are not related to the outcome variable
## and are therefore not included in the final dataset.

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# These features are dropped as they are not related to the outcome variable
# and are therefore not included in the final dataset.

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# These features are dropped as they are not related to the outcome variable
# and are therefore not included in the final dataset.
```

Figure 4: Drop unrelated features

```
## Drop Features by Correlation
## These features are dropped as they are highly correlated with other features
## and are therefore not included in the final dataset.

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# and are therefore not included in the final dataset.
```

Figure 5: Drop Features by Correlation

Following data preparation, we integrated the HOI values from the VDH_VA_HOI dataset into the ACS_VA dataset based on matching census tracts. This integration allowed us to train and validate a statistical machine learning model to predict the HOI accurately. The validation of this model provided insights into its predictive power and robustness.

The ACS_VA dataset now includes 1874 records and 54 characteristics, each of which has a correlation with the HOI of at least 0.25. We also have identified 223 missing values and have developed a strategy to compensate for these missing values with the median value of the corresponding column at the analysis stage.

A correlation of 0.25, while not strong, is sufficient to indicate statistical significance in our case, as we wanted to make sure that each selected variable had at least some relationship to the predicted HOI value during the variable selection process. We also tried a brute-force approach to determine which correlations would yield the best results. First, we divided the thresholds into 0.25, 0.50, and 0.75, and then tested the R-squared of each correlation through the overall models, and we found that 0.25 performed best over the other thresholds. Then, we also tried 0.30 and 0.20, which is plus or minus 0.05 for a correlation coefficient of 0.25, and the results also showed that 0.25 had the best performance. Therefore, we decided to use 0.25 as the correlation threshold.

It indicates that there is some relationship between the feature and the HOI, which may be nontrivial and worth considering in the model. Setting a lower threshold could result in a model that is too complex, with too many features that do not contribute significant information, leading to overfitting. A threshold of 0.25 helps to maintain a balance, keeping the model simpler and more interpretable without losing important information.

Encouraged by the model's performance on Virginia data, we extended our analysis to California. Using a similar approach, we collected the 2013-2017 ACS data for California, referred to as the ACS_CA dataset. We ensured that the features in the ACS_CA dataset precisely matched those in the ACS_VA dataset to maintain consistency in model application.

The final phase of our study involved applying the Virginia-trained model to the California dataset. This step tested the generalizability and adaptability of our model across different demographic and socioeconomic environments.

This structured approach to data handling not only ensured the reliability of our findings but also highlighted the potential for scalability of the HOI predictive model to other states, providing a robust framework for public health analytics.

4 Initial Model Evaluation

In our initial analysis using the VDH_VA_HOI dataset, we employed several supervised statistical machine learning models to predict the Health Opportunity Index (HOI) values, contrasting with the unsupervised methods used by the VDH team. The performance metrics of these models are depicted in Figure 6. While the results from the decision tree, random forest, and XGBoost models appeared reasonable, the Multiple Linear Regression (MLR) model exhibited an R-squared value of 1.0, which is highly unusual and suggests a perfect fit.

Given that the dataset met all standard assumptions for MLR—including the absence of multicollinearity—this perfect score raised concerns about possible data leakage or an inherent flaw in the dataset construction. A plausible explanation might be that the HOI, as constructed by the VDH team, includes variables that are highly inter-correlated or derivative of the inputs, leading to inflated correlations in our models.

To address these issues, we implemented dimensionality reduction through Principal Component Analysis (PCA) and shifted from using a Standard Scaler to a MinMax Scaler for feature normalization. Principal component analysis, or PCA, is a dimensionality reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set. PCA helps in reducing the dataset to its most informative features, thereby potentially alleviating problems related to high dimensionality and multicollinearity. The MinMax Scaler was chosen based on its appropriateness for the HOI's range of [0,1], aligning with the scaling method used by the VDH team. After applying PCA, we retained 75% of the data variance, which optimized model performance without overfitting. Figure 7 showcases the comparison of model metrics before and after these adjustments.

Our analysis demonstrated that XGBoost did not achieve the anticipated level of effectiveness on the VDH_VA_HOI dataset. This observation could stem from several pivotal factors intrinsic to XGBoost's operational mechanics and the dataset characteristics. XGBoost requires meticulous tuning of its hyperparameters. Suboptimal settings, such as the depth of trees or learning rate, may have hindered its ability to generalize effectively, leading to either underfitting or overfitting. The inherent complexity of XGBoost, while beneficial for capturing intricate patterns, may lead to overfitting, especially when the dataset lacks the requisite complexity or when noise is mistaken for signal.

```

Linear Regression:
MSE: 9.078078319381668e-22
MAE: 2.6623733514104703e-11
R^2: 1.0
-----
Decision Tree:
MSE: 0.0015003832001013269
MAE: 0.028861385743466667
R^2: 0.799581596702108
-----
Random Forest:
MSE: 0.0005027756505428863
MAE: 0.01574663027654666
R^2: 0.9328401617053169
-----
XGBoost:
MSE: 0.0003831920967878618
MAE: 0.014476227840028812
R^2: 0.9488139108799618
-----

```

Figure 6: VDH_VA_HOI with Standard Scaler

```

Linear Regression:
MSE: 0.0016260261903141237
MAE: 0.030832488282501277
R^2: 0.7827984392511842
-----
Decision Tree:
MSE: 0.0029629892395012895
MAE: 0.04148448204053333
R^2: 0.6042093964198083
-----
Random Forest:
MSE: 0.0015295330157802382
MAE: 0.028715108984274682
R^2: 0.7956878184230652
-----
XGBoost:
MSE: 0.0017876170538082936
MAE: 0.03160191098778056
R^2: 0.7612134316032434
-----

```

Figure 7: VDH_VA_HOI with PCA and MinMax Scaler

Random Forest, on the other hand, is less likely to overfit compared to XGBoost. Because it averages multiple deep decision trees, each trained on a different subset of the data, it generally provides a more generalized solution. Random Forest provides straightforward insights into feature importance based on how often features are used to split decisions in trees, which can be more intuitive in understanding which predictors are driving the HOI.

Our exploratory analyses affirm the feasibility of using supervised regression models like Random Forest for predictive tasks within public health data. This approach holds potential for broader application, promising robust, interpretable, and actionable insights that could significantly impact public health planning and interventions.

5 New Model Evaluation

In this phase of our analysis, we apply principal component analysis (PCA) to the ACS_VA dataset, opting for StandardScaler over MinMaxScaler due to the former's superior performance with raw data. Explorations indicated that MinMaxScaler, while effective in certain contexts, resulted in significantly lower performance when applied to the unprocessed attributes of the ACS data. PCA is utilized to reduce the dimensionality of the dataset, enhancing our model's ability to manage and interpret the vast array of socioeconomic and demographic features effectively. This technique not only simplifies the computational demands but also helps in isolating the most impactful features, thereby improving the robustness and accuracy of our predictive models, and normalizes and solves Linear regression issue. By reverting to StandardScaler, we aim to normalize the features within their natural range of values, maintaining the integrity and distribution of the raw data which is crucial for capturing the nuanced relationships within the social determinants of health.

5.1 Multiple Linear Regression

Multiple linear regression (MLR), is a common statistical method used to build a regression model between several explanatory variables and a response variable. This essentially is an extension of the ordinary least-squares regression (OLS) except that it involves more than one explanatory variable.

All assumptions for multiple linear regression are met by our dataset as: The observations are independent of each other, The variance of the errors is constant across all levels of the independent variables, and the residuals are normally distributed.

MLR is a simple model to implement and has little tuning process to do. The result model is intuitive and easy to interpret. Therefore if we would include this in our project. We aim to implement this model first to get a general sense of relationships between the explanatory and response variables.

MLR Equation based on ACS_VA (Rounded to keep three decimals):

$$\begin{aligned}
 y &= 0.422 \\
 &- 0.013 \times \% \text{ Households: Less than \$30,000} \\
 &+ 0.006 \times \% \text{ Population Age 18 to 64 for Whom Poverty Status Is Determined: Living in Poverty} \\
 &- 0.002 \times \% \text{ Population Age 18 to 64 for Whom Poverty Status Is Determined: At or Above Poverty Level} \\
 &- 0.018 \times \% \text{ Households: Less than \$100,000} \\
 &+ 0.008 \times \% \text{ Households: \$100,000 or More} \\
 &- 0.006 \times \% \text{ Households: \$150,000 to \$199,999} \\
 &- 0.010 \times \% \text{ Families: Income At or Above Poverty Level} \\
 &+ 0.005 \times \% \text{ Families: Income Below Poverty Level}
 \end{aligned}
 \tag{1}$$

The linear regression model reveals how various income brackets impact the outcome variable, HOI. It shows a negative relationship between lower income levels and HOI, with a greater negative impact observed for very low incomes, such as households earning less than \$10,000. Conversely, increases in poverty levels among the population aged 18 to 64 are associated with increases in HOI, possibly reflecting the activation of social supports or programs. Interestingly, higher income brackets, like those earning over \$100,000, generally promote positive changes in HOI, except in the \$150,000 to \$199,999 range where the effect is negative, suggesting complex economic dynamics at higher income levels. Additionally, families above the poverty line show a negative correlation with HOI, indicating reduced reliance on the outcomes measured, while those below the poverty line show a positive correlation, likely due to increased need or eligibility for aid.

5.2 Decision Tree

Decision tree is a supervised machine-learning model commonly used for classification and regression. It converts the variables into flowchart-like tree structures of decisions and possible outcomes. Each node would represent a variable in the data, the edges would represent possible decision rules and the leaves represent the corresponding outcomes.

All assumptions for the decision tree are met by our dataset: The data can be partitioned into subsets based on the input variables derived from the data features.

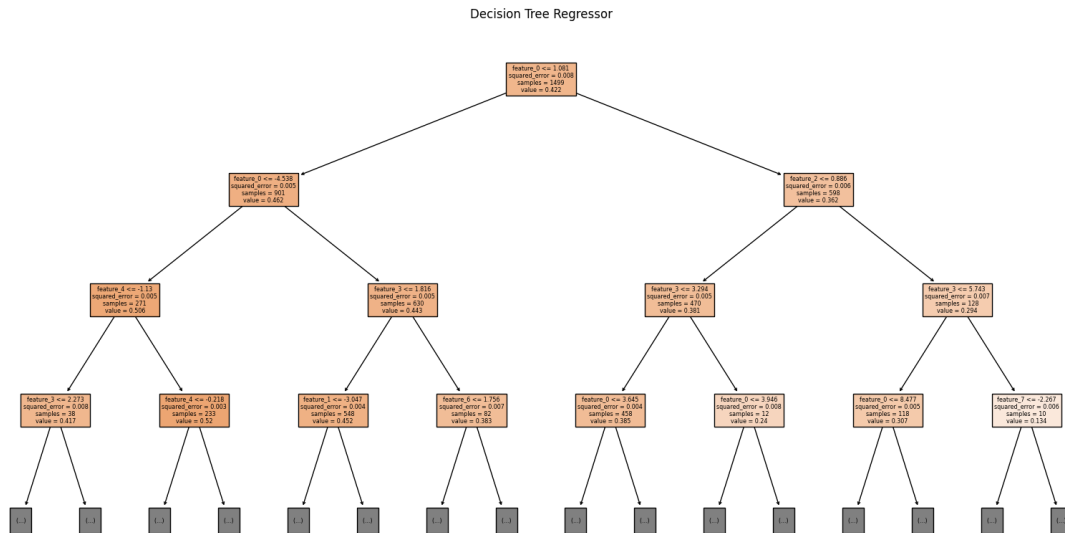


Figure 8: Decision Tree visualization.

The Decision Tree Regressor plot illustrates how the decision-making process regarding a target variable is influenced by household income and poverty status. Beginning at the root node, the model evaluates the percentage of households earning less than \$30,000. Based on this value, it branches into further decisions, analyzing more specific income

categories like households earning \$100,000 or more and those less than \$10,000. Each node in the tree represents a decision point where the dataset is split into increasingly homogeneous subsets based on income thresholds, poverty levels among the 18 to 64 age group, and families above or below poverty level. The decision tree methodically segments the population to pinpoint patterns and interactions between various income brackets, effectively capturing complex relationships between socioeconomic factors and the target outcome. This approach provides a structured analysis of how different socioeconomic strata could influence the targeted outcome, facilitating nuanced insights into the data's underlying structure and guiding policy formulation or targeted interventions.

5.3 Random Forests

The random forest algorithm is based on the decision tree model. It is made up of a collection of decision trees, where each tree is comprised of a bootstrap sample from the population. Each tree is then developed individually, but instead of considering all the features when splitting a node, it uses a specified random subset of features. Each tree makes the regression prediction and then aggregates the results.

Random Forests inherit the assumptions of decision trees. Additionally, Random Forests assume that the individual decision trees in the ensemble are diverse, meaning that they capture different aspects of the data. This diversity is achieved through bootstrap sampling and random feature selection. Our data meets these assumptions.

As it takes the mean of all the decision trees, Random Forest is very robust to outliers. There is not a large amount of tuning compared to XGBoost and the prediction results could also be accurate. So we decided to include this.

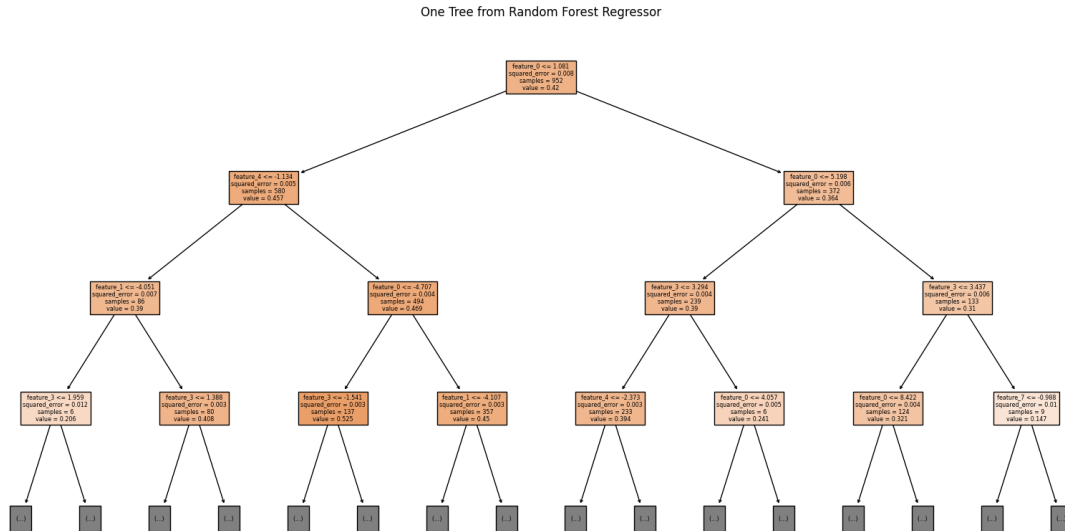


Figure 9: Random Forest Visualizaiton

This plot illustrates a single tree from a Random Forest Regressor, representing one of many such trees in the ensemble that collectively determine a target variable. The tree begins with a root node using Feature 4 (Households earning \$100,000 or more) to make the initial split, suggesting the significant impact of this income level on the target outcome. It then branches based on various other socio-economic features, such as Feature 0 (Households earning less than \$30,000) and Feature 3 (Households earning less than \$10,000), each node refining the prediction by segmenting the data into subsets that share similar characteristics and outcome values. Notably, the tree utilizes thresholds for these features to create splits, maximizing the homogeneity of the target variable within each group. This approach allows the Random Forest model to capture complex interactions between multiple features, improving prediction accuracy by aggregating insights from multiple decision paths.

5.4 XGBoost

XGBoost is an ensemble machine-learning algorithm that optimizes the accuracy of predictions using gradient boosting. It sequentially adds decision trees to the overall model, each one attempting to correct the errors from the previous one.

No specific distributional assumptions are made in XGBoost.

The XGBoost model incorporates regularization terms, therefore it prevents overfitting. Theoretically and from many previous works, XGBoost also gives the highest prediction accuracy among the models. Therefore we consider it as a valid model to use in our project.

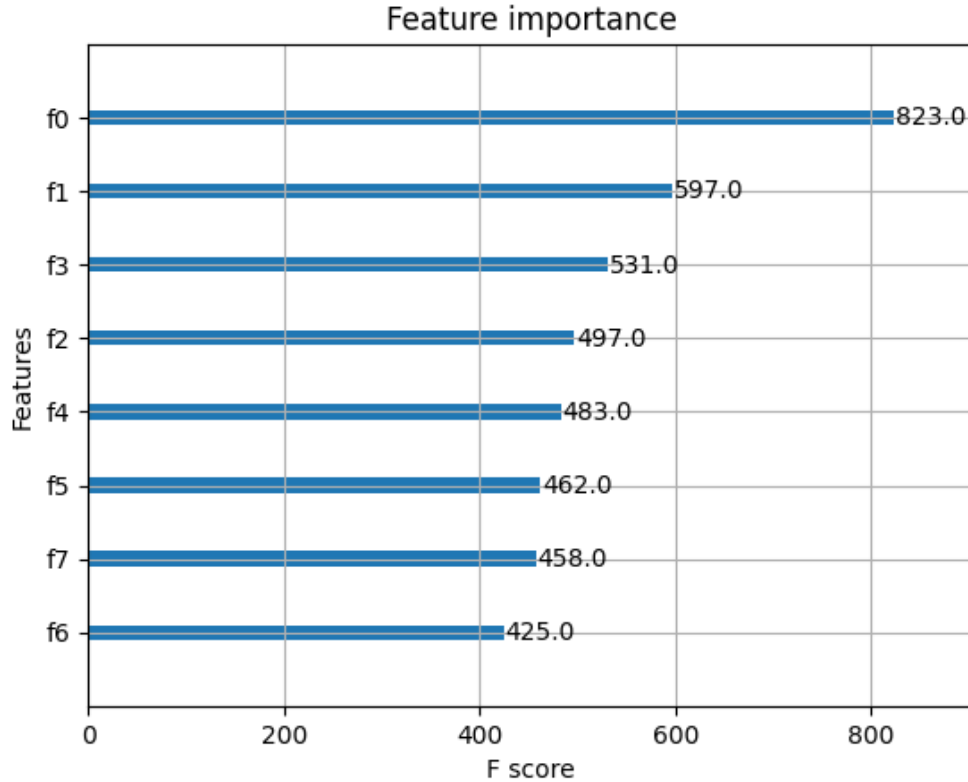


Figure 10: XGBoost Top 10 Results

This plot from an XGBoost model illustrates the importance of various features based on their F-scores, which measure the number of times a feature is used to split the data across all trees. Feature f0 (% Households: Less than \$30,000) is the most significant, showing the highest F-score of 823, indicating it plays a crucial role in the model's decision-making process. The following features, f1 through f7 (check the appendix), also contribute significantly but to a lesser extent, with their importance decreasing in descending order. This hierarchy of feature importance helps in understanding which variables most influence the model's predictions, aiding in focusing efforts on the most impactful factors for further analysis or data collection.

5.5 Model Comparison

In our comprehensive assessment of different statistical models, we observed that Random Forests outperformed other models significantly. The Random Forest model exhibited the lowest Mean Squared Error (MSE) and the second lowest Mean Absolute Error (MAE), alongside the highest R-squared value of approximately 60.36%, indicating that it explains about 60.36% of the variance in our target variable. This model's robust performance underscores its efficacy in capturing the complex relationships within our data.

Conversely, Linear Regression, while not the best, still performed commendably with the second lowest MSE and the lowest MAE, achieving an R-squared value of 58.10%. This suggests a substantial proportion of variance in the target variable is accounted for by the Linear Regression model, marking it as a strong contender for accurate predictions.

The Decision Tree model lagged behind, registering the highest MSE and MAE, and an R-squared value of only 28.99%. This relatively poor performance indicates a weaker fit to the data compared to other models.

XGBoost, although surpassing the Decision Tree in performance, yielded the third lowest MSE and MAE and held moderate explanatory power with its R-squared value, which did not approach those of the more effective Random Forest and Linear Regression models. Therefore, we will use Random Forest Model to predict HOI value for other states.

```

Linear Regression:
MSE: 0.0030472833040705786
MAE: 0.04255869364802291
R^2: 0.5899645046812778
-----
Decision Tree:
MSE: 0.004983331722412284
MAE: 0.05618769997733333
R^2: 0.32945424260116773
-----
Random Forest:
MSE: 0.0029459680235283158
MAE: 0.04295035840837867
R^2: 0.6035972578896877
-----
XGBoost:
MSE: 0.003368359885843405
MAE: 0.045974816140525794
R^2: 0.5467611717103655
-----

```

Figure 11: ACS_VA Dataset Model Comparison

6 Results

We employed our random forests model due to its superior predictive performance relative to other methods, aiming to predict the HOI for census tracts in California. However, as California lacks data on HOI scores, we needed an alternative validation approach for our model.

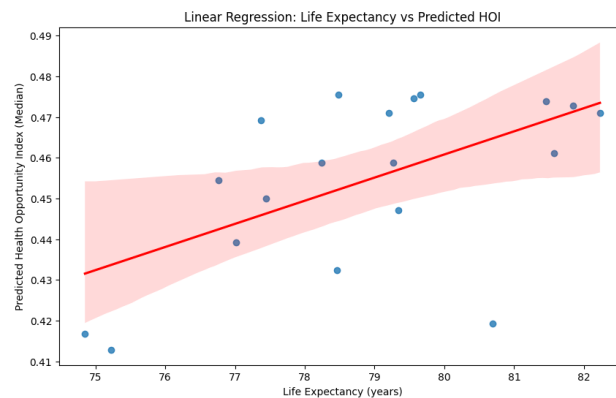


Figure 12: ACS_CA's Positive Linear Relationship between Predicted HOI and Life Expectancy

To validate our California models using ACS data for predicting Health Opportunity Index (HOI) in census tracts, we utilized life expectancy as a proxy measure. Human life expectancy is a statistical measure of the estimate of the average remaining years of life at a given age. The most commonly used measure is life expectancy at birth. We collect the life expectancy data from the Institute for Health Metrics and Evaluation (IHME) which is an independent global health research center at the University of Washington. With life expectancy data available only at the county level in California, our analysis focused on the 20 counties encompassing the available census tracts. Our hypothesis suggested a positive linear relationship between predicted HOI scores and life expectancy.

Confirming our hypothesis, we computed the median HOI score for each county and observed a correlation coefficient of approximately 0.5723 between the average HOI score and life expectancy. This finding, alongside an R-squared value of 0.328 and a low p-value of 0.0104 for the F-statistic, provided strong evidence against the null hypothesis, indicating a statistically significant positive relationship between life expectancy and the Predicted Health Opportunity Index.

To further validate the use of life expectancy as a metric, we applied the same approach to verify our Virginia models, previously validated using available HOI data. Our analysis revealed a correlation coefficient of 0.83, an R-squared value of 0.6949, and a low p-value of 2.69e-6, supporting the positive linear relationship between life expectancy and HOI scores.

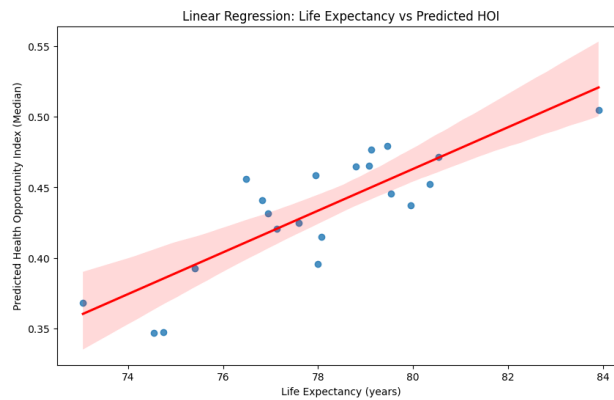


Figure 13: ACS_VA's Positive Linear Relationship between Predicted HOI and Life Expectancy

In summary, our endeavor to validate California models using ACS data showcased the effectiveness of employing life expectancy as an alternative metric for assessing predictive capability. Despite the absence of direct HOI data in California, our methodology provided valuable insights, highlighting the potential applicability of our approach in estimating HOI scores, even in regions lacking direct data. Moving forward, refining and validating our models could enhance their utility across diverse contexts, facilitating more informed decision-making in public health policy and resource allocation.

OLS Regression Results						
Dep. Variable:	Predicted_Health_Opportunity_Index	R-squared:	0.328			
Model:	OLS	Adj. R-squared:	0.288			
Method:	Least Squares	F-statistic:	8.282			
Date:	Sun, 05 May 2024	Prob (F-statistic):	0.0104			
Time:	08:11:22	Log-Likelihood:	50.630			
No. Observations:	19	AIC:	-97.26			
Df Residuals:	17	BIC:	-95.37			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0062	0.156	0.040	0.969	-0.322	0.335
life expectancy	0.0057	0.002	2.878	0.010	0.002	0.010
Omnibus:	4.384	Durbin-Watson:	2.440			
Prob(Omnibus):	0.112	Jarque-Bera (JB):	2.485			
Skew:	-0.825	Prob(JB):	0.300			
Kurtosis:	3.561	Cond. No.	3.01e+03			

Figure 14: ACS_CA OLS Summary

OLS Regression Results						
Dep. Variable:	Predicted_Health_Opportunity_Index	R-squared:	0.695			
Model:	OLS	Adj. R-squared:	0.679			
Method:	Least Squares	F-statistic:	43.27			
Date:	Mon, 06 May 2024	Prob (F-statistic):	2.69e-06			
Time:	00:49:14	Log Likelihood:	49.284			
No. Observations:	21	AIC:	-94.57			
Df Residuals:	19	BIC:	-92.48			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-0.7170	0.175	-4.099	0.001	-1.083	-0.351
Life expectancy	0.0148	0.002	6.578	0.000	0.010	0.019
Omnibus:	1.084	Durbin-Watson:	1.666			
Prob(Omnibus):	0.581	Jarque-Bera (JB):	0.795			
Skew:	-0.052	Prob(JB):	0.672			
Kurtosis:	2.053	Cond. No.	2.57e+03			

Figure 15: ACS_VA OLS Summary

7 Considerations and Limitations

In addition to evaluating various modeling approaches, we are aware of inherent limitations with our existing datasets. One significant constraint is the inability to verify the robustness of our methods using data from other time periods. We hypothesize that there likely would not be major changes in the data trends from year to year, and thus anticipate that our model should generally perform well on datasets from different years. But we don't have the empirical evidence to assert this with certainty, especially considering the possibility of influential pandemic outbreaks that can cause significant changes in the citizens well-beings and therefore the dataset. This limitation underscores the need for caution when generalizing the findings derived from the 2017 dataset to other years.

Another challenge we face with our dataset is the concern of our model predictions. Despite the mean squared error (MSE) is relatively low (below 0.05 for all models), the R-squared value is not particularly high (highest value for random forest being 0.6). This suggests that our model does not account for a significant portion of the variance in the predicted Health Opportunity Index (HOI). This indicates that our models have limited explanatory power. Enhancing the accuracy of the R-squared value would provide a more reliable measure of how well our predictions match the actual data, and would likely further reduce the test MSE. To improve the model's performance we can try to further tune the model. This could lead to a more robust model that can explain a greater proportion of the variance in the HOI, thereby increasing the overall predictive precision.

Our efforts to improve the robustness of the predictive accuracy of the model depended heavily on access to the most recent dataset. The Virginia Department of Health team compiled the most recent Virginia Health Opportunity Index using raw data from the 2020 American Community Survey (ACS), which is a key resource in this regard. Incorporating this dataset was not only complementary but transformative, with the potential not only to validate our model but also to calibrate it against more modern measures of health opportunity. Our initial communication with Mr. Rexford Anson-Dwamena, the VDH HOI project leader, was promising. An understanding was reached that our team would be granted access to the dataset, a move that expressed support for our shared goals of academic inquiry and advancing public health intelligence. However, despite numerous attempts to solidify this agreement and gain access to the dataset, our communications have been unsuccessful. We have been reaching out to him through various channels but still waiting for responses.

8 Impact and Future Work

The real-world applications of using statistical machine learning models to predict the Health Opportunity Index (HOI) extend beyond Virginia to have broader implications for public health research and practice. Leveraging readily available and accessible data, such as that collected by the American Community Service (ACS), can streamline the prediction process for HOI in Virginia and other states. It facilitates the development of nationwide evidence-based policy recommendations aimed at improving overall public health outcomes. By targeting interventions towards areas identified as most disadvantaged, such as those with lower predicted HOI values, communities facing health disparities can receive the necessary support and resources to enhance their well-being. In addition, by using existing datasets, researchers and practitioners can minimize the need for extensive data collection efforts, reducing costs and time associated with model development. Additionally, statistical machine learning models can be integrated into public health surveillance systems to continuously monitor changes in HOI over time. This enables early detection of emerging health disparities and allows for timely interventions to mitigate their impact on population health.

Due to the limited availability of datasets, our project is based on data from 2017, which could be dated from now. Given the 7-year time period and the global outbreak of the COVID-19 pandemic, there might be significant changes in health opportunities during the period. Thus, the possible next step for our project is to collect more recent data for analysis and come up with the most updated prediction for HOI.

9 Conclusion

Throughout the analysis stage, we initially faced a surprisingly high R-square value of 1 when building the linear regression model for the original Virginia HOI dataset (VDH_VA_HOI). After resolving the issue by normalizing the variables using the Min-Max scaler and PCA, we rebuilt the models and discovered the potential for using supervised statistical machine learning model for HOI prediction in Virginia.

Then, after collecting a new ACS dataset (ACS_VA), we trained various machine learning models for predicting HOI in Virginia, including Multiple Linear Regression, Decision Tree, Random Forests, and XGBoost. After comparing the effectiveness of these models in predicting HOI using metrics like mean squared error, R-squared, and mean absolute

error, we concluded that the random forest model stood out as the most effective one. This determination was based on its notably lower MSE and higher R squared compared to the other models assessed.

In the final stage, we tested the predictive power of the random forest model towards California state. Using the self-collected California ACS dataset (ACS_CA), We discovered that there is a positive linear relationship between life expectancy and the predicted HOI in California. It further verified the predictability of HOI with our model as areas predicted as having higher HOI are associated with longer life expectancy.

As we move forward, this research paves the way for further exploration and refinement of predictive models, ultimately contributing to more informed decision-making and targeted interventions in public health policy and resource allocation.

10 Appendices

Feature Index	Feature Name
0	% Households: Less than \$30,000
1	% Population Age 18 to 64 for Whom Poverty Status Is Determined: Living in Poverty
2	% Population Age 18 to 64 for Whom Poverty Status Is Determined: At or Above Poverty Level
3	% Households: Less than \$10,000
4	% Households: \$100,000 or More
5	% Households: \$150,000 to \$199,999
6	% Families: Income At or Above Poverty Level
7	% Families: Income Below Poverty Level
8	% Population Under 18 Years of Age for Whom Poverty Status Is Determined: At or Above Poverty Level
9	% Population Under 18 Years of Age for Whom Poverty Status Is Determined: Living in Poverty
10	Gini Index
11	% Households: \$10,000 to \$14,999
12	% Households: \$125,000 to \$149,999
13	% Total: with Health Insurance Coverage: Private Health Insurance
14	Median Family Income (In 2017 Inflation Adjusted Dollars)
15	% Households: \$20,000 to \$24,999
16	% Households: \$15,000 to \$19,999
17	Renter-Occupied Housing Units
18	% Renter-Occupied Housing Units: Less than \$1,000
19	% Households: \$100,000 to \$124,999
20	% Households: \$200,000 or More
21	% Civilian Population in Labor Force 16 Years and Over: Employed
22	% Civilian Population in Labor Force 16 Years and Over: Unemployed
23	% Population Age 65 and Over for Whom Poverty Status Is Determined: At or Above Poverty Level
24	% Population Age 65 and Over for Whom Poverty Status Is Determined: Living in Poverty
25	Average Household Size
26	% Population 16 Years and Over: in Labor Force: Civilian: Employed
27	% Total: with Health Insurance Coverage: Public Health Coverage
28	% Total: with Health Insurance Coverage
29	% Total: No Health Insurance Coverage
30	% Households: \$30,000 to \$34,999

Figure 16: Appendix 1

31	% Households: \$25,000 to \$29,999
32	% Renter-Occupied Housing Units: Householder 15 to 24 Years
33	% Population 16 Years and Over: Not in Labor Force
34	% Population 16 Years and Over: in Labor Force
35	% Households: \$35,000 to \$39,999
36	% Population 16 Years and Over: in Labor Force: Civilian
37	Families:
38	County
39	Census Tract
40	% Renter-Occupied Housing Units: \$2,000 to \$2,499
41	% Renter-Occupied Housing Units: Less than \$300
42	% Renter-Occupied Housing Units: \$500 to \$799
43	% Population 16 Years and Over: in Labor Force: Civilian: Unemployed
44	% Renter-Occupied Housing Units: \$1,000 to \$2,499
45	% Renter-Occupied Housing Units: \$2,500 to \$2,999
46	% Renter-Occupied Housing Units: \$300 to \$499
47	% Households: \$40,000 to \$44,999
48	% Renter-Occupied Housing Units: Householder 35 to 44 Years
49	Population Density (Per Sq. Mile)
50	% Renter-Occupied Housing Units: Householder 45 to 54 Years
51	% Households: \$75,000 to \$99,999
52	% Renter-Occupied Housing Units: \$800 to \$999

Figure 17: Appendix 2